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Emotional intelligence and drawing inferences from nonverbal cues in sports

Philip Furley^a, Geoffrey Schweizer^b and Sylvain Laborde^c

^aInstitute for Training and Computer Science in Sport, German Sport University, Cologne, Germany;

^bDepartment of Sport and Exercise Psychology, University of Heidelberg, Heidelberg, Germany; ^cInstitute of Psychology, German Sport University Cologne, Cologne, Germany

ABSTRACT

We aimed to investigate whether emotional competencies (EC) correlate with the ability to judge or predict the score in sports competitions based on thin slices of nonverbal emotional expressions. We created a German version of the Profile of Emotional Competence (PEC; Brasseur, S., Grégoire, J., Bourdu, R., & Mikolajczak, M. (2013). The Profile of Emotional Competence (PEC): Development and validation of a measure that fits dimensions of Emotional Competence theory. *PLoS ONE*, 8(5), e62635) and assessed its psychometric properties. In Study 1, 379 participants completed the PEC and rated whether videotaped athletes were leading or trailing. We hypothesised that the ability to distinguish between leading and trailing athletes was predicted by participants' interpersonal, but not intrapersonal EC. In Study 2, 310 participants completed the PEC, a test assessing individual differences in emotion recognition competence (ERI; Scherer, K. R., & Scherer, U. (2011). Assessing the ability to recognize facial and vocal expressions of emotion: Construction and validation of the emotion recognition index. *Journal of Nonverbal Behavior*, 35(4), 305–326), and predicted the performance of darts players based on their nonverbal behaviour. In Study 1, we found small correlations between intrapersonal EC and accuracy of score estimates, but no correlations between interpersonal EC and accuracy of score estimates. Contrary to study 1, in study 2, intraindividual EC correlated neither with accuracy of performance predictions nor with ERI scores, whereas interpersonal EC correlated with ERI scores. Our results suggest that further evidence for the validity of EC measures in the context of sports is needed.

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Most research in sports treats emotions as transient states (Furley, 2021b; Ruiz & Robazza, 2021). However, it is also possible to investigate more stable aspects of how individuals deal with emotional experiences in sport. The most prominent concept reflecting these relatively stable emotional characteristics has been termed emotional intelligence (EI) (Goleman, 1995; Petrides & Furnham, 2003; Salovey & Mayer, 1990). EI has been proposed to be a construct that reflects the way people deal with their own emotions (*intrapersonal*) and the emotions of others (*interpersonal*). Research in sports has shown a positive

correlation between EI and performance in several sports domains (Kopp & Jekauc, 2018; Laborde et al., 2016). Hence, there has been a growing interest in measuring how well athletes, officials, and coaches are able to deal with their own and others' emotions. While there have been increasing studies investigating the relationship between measures of EI and performance, fewer studies have investigated into the *interpersonal* aspect of EI in sports (Laborde et al., 2016). One reason for this is that popular measurement tools of EI like the Mayer-Salovey-Caruso Emotional Intelligence Test (Mayer et al., 2002) do not explicitly distinguish between *intra*- and *interpersonal* dimensions. Hence, the goal of the present research was to test if an economic, time-efficient self-report measure of *intra*- and *interpersonal* emotional competencies (Profile of Emotional Competence: PEC; Brasseur et al., 2013) was predictive of interpreting emotional displays of athletes during competition.

The PEC (Brasseur et al., 2013) was built on the premise that the most important emotional competencies (EC) that contribute to EI must be subdivided into *interpersonal* and *intrapersonal* emotional skills. *Intrapersonal* emotional skills entail how well individuals identify, express, understand, regulate, and use their own emotions, whereas *interpersonal* emotional skills assess how well individuals identify, listen, understand, regulate, and use the emotions of others. Although Brasseur and colleagues show that the PEC has promising psychometric properties, including good internal consistency of the 10 subscales, they also acknowledge that further studies are needed to provide concurrent validity, in particular regarding the distinction between *intrapersonal* EC and *interpersonal* EC.

Although researchers acknowledge the importance of validating measures in psychology, it has recently become evident that in practice psychological scientists give very little attention to validating their measures (Eronen & Bringmann, 2021) and have even spoken of a validation crisis in psychology (Schimmack, 2010; Schimmack, 2019). In this respect, a recent review found that a random representative sample of articles published in leading psychological journals reported no validity evidence of the constructs used (Flake et al., 2017). One reason for this state of affairs is that providing validity information for a psychological construct is very difficult, time consuming, and is not adequately incentivized (e.g., by journal editors; Eronen & Bringmann, 2021; Flake et al., 2017). As the construct of EI is gaining momentum in applied fields like sport psychology (Kopp & Jekauc, 2018; Laborde et al., 2016) although research shows that this construct is problematic to measure and lacks construct validity (e.g., Elfenbein & MacCann, 2017; Laborde & Allen, 2016; Olderbak et al., 2019), the present research attempts to advance construct validity for the *interpersonal* subscale of the PEC (Brasseur et al., 2013).

Given the emotional nature of professional sport competitions, sports appear to be an ideal context for investigating *interpersonal* EC as athletes are constantly sending out emotional nonverbal signals (Furley, 2019; Furley, 2021a; Furley 2021b) and accuracy of inferences can be assessed based on the objective score criterion (Furley & Schweizer, 2014). Drawing inferences from brief observations of an athlete's nonverbal behaviour (NVB) during a game clearly falls within the conceptualisation of *interpersonal* EC as the observer is trying to identify and understand the emotional state another person is currently experiencing (Brasseur et al., 2013). Hence, we hypothesised that *interpersonal* EC (and not *intrapersonal* EC) should positively predict how well people can draw inferences from emotional expressions they perceive in others. We tested this hypothesis by correlating participants' EC scores with their accuracy in judging the current score during televised

sports competitions without being aware of the score (Experiment 1) and predicting performance tendencies based on an athlete's preperformance NVB (Experiment 2).

Study 1

Method

Participants

379 participants (193 male and 185 female; $M_{age} = 24.9$ years; $SD = 4.4$ years, one participant did not indicate gender) completed both the German version of the PEC and the score estimation task. An additional 180 participants (77 male, 103 female; $M_{age} = 24.2$, $SD = 5.4$) completed only the German version of the EC questionnaire for psychometric analyses. The large majority of participants had no particular sports expertise and were college students enrolled at German universities, primarily in sport science or psychology majors. All participants participated voluntarily without compensation and were recruited within research methodology classes and bachelor and master's projects. Participants gave written informed consent before the experiment. An a priori sample size calculation with G*Power (Faul et al., 2007) showed that the study had sufficient power (.95) to detect correlation coefficients of $r = .20$ on two-tailed Pearson correlations. A sample size of 314 would be sufficient to achieve 95 percent power.

Profile of Emotional Competence

For the assessment of EC we back-translated (Brislin, 1970) the full English version (50 items) of the Profile of Emotional Competence (PEC; Brasseur et al., 2013) to German (see online supplement for translated questionnaire). The PEC allows measuring the intra-personal and interpersonal emotional competences separately for the identification, understanding, expression, regulation and use of emotions distinctly for one's emotions and others' emotions. Psychometrics for this measure have been assessed in several studies with a total sample size of nearly 22.000 subjects (Brasseur et al., 2013).

The PEC consists of 50-items that comprise ten subscales with two factors (intra- and interpersonal EC) and one global EI score. The ten subscales are: Identify my emotions; identify others' emotions; understand my emotions; understand others' emotions; express my emotions; listen to others' emotions; regulate my emotions; regulate others' emotions; use my emotions; use others' emotions.

Descriptive statistics of the German version of the PEC can be seen in Table 1 (for the entire sample of 559 participants). Further, to check whether we could replicate the original factor structure we ran a Confirmatory Factor Analysis (CFA) with AMOS 17 on our sample of 559 participants. We compute the χ^2 statistic as a subjective index of fit, with large chi-square values relative to degrees of freedom indicating a poor fit, and small values indicating a good fit (Jöreskog, 1993). As χ^2 is quite sensitive to sample size and inflates with large samples (Raykov, 1998) we further computed the comparative fit index (CFI), the Tucker-Lewis index (TLI), the root mean square of approximation (RMSEA), the standardised root mean square residual (SRMR). Values between .90 and .94 indicate acceptable fit, whereas values of .95 and higher indicate a relatively good fit for the CFI and the TLI (Hu & Bentler, 1999). Values smaller than .08 indicate acceptable fit for the RMSEA, values smaller than .05 indicate a good fit. A good fit is indicated by values smaller than .08 for SRMR (Hu & Bentler, 1999).

Table 1. Descriptive statistic Profile of Emotional Competences – German validation.

	<i>M</i>	<i>SD</i>	Cronbach's alpha
Identification (own)	3.63	0.67	.72
Identification (other)	3.69	0.63	.75
Understanding (own)	3.59	0.77	.71
Understanding (others)	3.71	0.63	.72
Expression (own)	3.38	0.68	.64
Expression (others)	3.93	0.72	.73
Regulation (own)	3.08	0.70	.71
Regulation (others)	3.57	0.56	.69
Use (own)	3.48	0.60	.68
Use (others)	3.20	0.68	.73
Intrapersonal Emotional Competences	3.43	0.45	.86
Interpersonal Emotional Competences	3.62	0.48	.88
Global Emotional Competences	3.52	0.42	.91

As a model for the CFA, we took the ten subscales and the two main factors of the PEC to see whether the original structure would replicate in the German sample. The first CFA indicated acceptable fit to the data: $\chi^2 = 199.68$, $df = 34$, $p < .001$, CFI = .91, TLI = .89, RMSEA = .09, SRMR = .06. However, analysis of the modification indices (M.I.) suggested that correlating the error variance between two subscales from the interpersonal factor, namely “regulating others emotions” and “using others emotions” would improve model fit (value modification index = 74.28). Thus, our final CFA model indicated a good fit to the data supporting the PEC factor structure in a German sample: $\chi^2 = 120.21$, $df = 33$, $p < .001$, CFI = .95, TLI = .94, RMSEA = .07, SRMR = .05.

Regarding the internal reliability of the PEC subscales, all subscales show acceptable reliability ($\alpha > .70$) except three: expression of own emotions ($\alpha = .64$), regulation of others' emotions ($\alpha = .69$) and use of own emotions ($\alpha = .68$). Given the sample of the current study was mainly constituted of sport sciences students, future data collection has to be realised with a less specific sample. While checking whether the data from the German sample was matching the original structure of the PEC in 10 subscales and 2 factors, it appeared necessary to correlate the error variance of two dimensions from the interpersonal factor, “regulating others' emotions” and “using others' emotions”. According to Brasseur and colleagues (2013), the first dimension refers to being able to regulate stress or emotions in others in a context-appropriate manner. The second refers to being able to use others' emotions to help them improve reflection, decisions and actions. Even though both competences are conceptually distinct; we observed some overlap between them, in that using others' emotions could be seen as regulating others' emotions with the objective to achieve a specific goal.

In summary, both the internal validity and factor structure of the German version of the PEC appear acceptable, and would advocate for its use in the current study. However, further research should investigate further convergent, divergent, and criterion validity, as well as test-retest reliability of the German PEC.

Estimating the score in televised sports competitions

In order to test the predictive validity of the German PEC we measured how well people (sample of 379 participants) could estimate the score during sports contests based on

short videos of athletes' body language (NVB) with the score occluded. We used the identical task to Furley and Schweizer (2014). In this task participants viewed a sample of videos showing professional basketball or table tennis players and had to estimate if the depicted athletes were leading or trailing on a continuous 100-point scale. Videos were randomly selected from five different score categories: far behind, close behind, draw, close lead, and high lead. E-prime 2.0 professional was used to collect the responses of the participants. The software randomly selected videos from the score categories for both basketball and table tennis videos. Each participant rated 72 videos in a different random order by moving a mouse cursor from the middle of the continuous scale to either the "high-lead" or "far-behind" pole. The full details of this task can be found in Furley and Schweizer (2014).

Preliminary study

To verify that the videos in the five score categories differed in terms of emotional content we conducted a preliminary study in which participants were asked to judge the sports videos in terms of emotional valence (positive vs. negative) and activation level (high vs. low) on the sports affect grid (Woodman et al., 2009). This measurement tool is presented as a 9×9 grid: the vertical axis evaluates activation (from extremely low to extremely high) and the horizontal axis evaluates emotional valence (from negative feeling to positive feeling).

40 participants ($M_{\text{age}} = 33.7$; $SD = 12.7$; 20 female) rated the 100 table tennis videos (with the score occluded) in random order on emotional valence and emotional activation. A 1×5 (far behind, close behind, draw, close lead, and high lead) ANOVA revealed a significant main effect of score category on overall emotional valence ratings, $F(4, 156) = 21.58$, $p < .001$, $\eta^2_p = .36$, indicating that perceivers rated trailing table tennis players as displaying more negative emotions and leading athletes as displaying more positive emotions. Polynomial linear contrast analysis showed a strong linear relationship between the emotional valence ratings and the score categories, $F(1, 39) = 31.76$, $p < .001$, $\eta^2_p = .45$. No significant effects were evident on emotional intensity across the score categories for the table tennis videos $F(4, 156) = 2.10$, $p = .100$, $\eta^2_p = .05$.

40 different participants ($M_{\text{age}} = 29.6$; $SD = 10.2$; 20 female) rated the 100 basketball videos (with the score occluded) in random order on emotional valence and emotional activation. A 1×5 (far behind, close behind, draw, close lead, and high lead) ANOVA revealed a significant main effect of score category on overall emotional valence ratings, $F(4, 156) = 32.28$, $p < .001$, $\eta^2_p = .45$, indicating that perceivers rated trailing basketball players as displaying more negative emotions and leading athletes as displaying more positive emotions. Polynomial linear contrast analysis showed a strong linear relationship between the emotional valence ratings and the score categories, $F(1, 39) = 94.74$, $p < .001$, $\eta^2_p = .71$. No significant effects were evident on emotional intensity across the score categories for the basketball videos $F(4, 156) = 0.94$, $p = .441$, $\eta^2_p = .02$.

In conclusion, the preliminary studies showed that trailing table tennis and basketball players display negative emotional states in their NVB, whereas leading athletes showed positive emotional states in their NVB. Therefore, the videos can be considered suitable to test the hypothesis that *interpersonal EC* should be predictive of drawing accurate inferences from the emotional displays of athletes.

Procedure

Participants first filled out a questionnaire gathering demographic data. Half the participants first completed the PEC and then the score estimation task. The other half first did the score estimation task and then the PEC. Participants were instructed to answer as accurately as possible. Speed was not emphasised. Every perceiver was tested individually.

Data analyses

A 2 (basketball vs. table tennis) $\times$ 5 (far behind, close behind, draw, close lead, and high lead) repeated measures ANOVA on the average score estimates within the respective experimental categories was run to test how well participants could estimate the current score based on the NVB shown in the videos. We followed up this ANOVA with a series of polynomial linear contrast analyses. In order to analyze individual difference in any potential overall effects, we computed two variables for the basketball and table-tennis videos respectively as an index of how well participants were able to estimate the score of the observed athletes. For the first individual difference variable we simply subtracted the average scores in the far-behind-category from the average score in the high-lead-category. This variable was labelled *difference*. Higher differences in this variable indicate better performance in the score estimation task as participants were better able to distinguish between athletes who were currently far behind and athletes who were leading by far. We computed a further variable that included the means of the close-lead, draw, and close-behind categories as a potentially more sensitive individual difference variable in score-estimation performance. To this end we calculated the slope of the regression line of the five score categories for every participant using the below formula. This variable was labelled *trend*. The x-values for every participant were set as 1 for far behind, 2 for close behind, 3 for draw, 4 for close lead, and 5 for high lead. The corresponding y-values were the mean scores of the participants in the respective categories.

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Again lower values for the regression slope indicate poorer performance on the score estimation task and higher values indicate better performance. To test for an association between the different EC as measured with the PEC and the performance on the score estimation task we ran a series of two-tailed Pearson correlations and tested them against each other.

Results

Figure 1 visualises the descriptive results of Experiment 1. ANOVA revealed a significant main effect of sport on overall score estimates ($F[1, 378] = 90.82, p < .001, \eta^2_p = .19$). This was mainly explained by overall higher ratings (closer to the “high lead” pole) in the table tennis videos. Further, there was a significant main effect of actual score on the score estimates ($F[3.746, 1415.969] = 212.51, p < .001, \eta^2_p = .36$) as score estimates

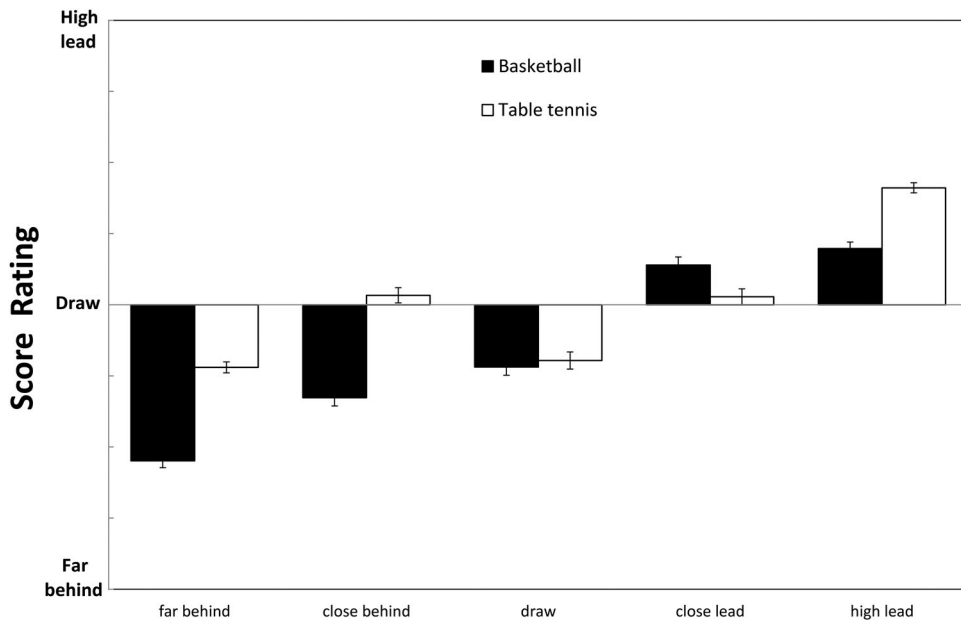


Figure 1. Mean score (0 = far behind; 1 = high lead) estimates as a function of score category and sport. Error bars represent standard errors.

corresponded in a linear manner with the score categories in both the table tennis and basketball videos. A significant interaction between sport and actual score ($F[3.517, 1329.422] = 41.75, p < .001, \eta^2_p = .10$) was also evident.

Follow-up polynomial linear contrasts across both sports revealed a strong linear relationship between the score estimates and the score categories ($F[1, 378] = 639.70, p < .001, \eta^2_p = .63$) demonstrating that the score estimates corresponded in a linear manner with the scores during the game. The significant interaction in the overall ANOVA is explained by the larger effect size in the linear contrast analysis for the basketball stimuli ($F[1, 378] = 549.29, p < .001, \eta^2_p = .59$) compared to the table tennis stimuli ($F[1, 378] = 356.70, p < .001, \eta^2_p = .49$).

Individual differences in estimating the score based on athletes' NVB

Table 2 shows that all four individual difference variables in the score estimation task (difference and trend for both basketball and table tennis videos) were significantly correlated. In contrast to our hypothesis, the interpersonal EC score was not significantly correlated with any of the individual difference variables in the score-estimation task. However, the intrapersonal EC score was significantly correlated—albeit weakly—with all four individual difference variables on the score estimation task.

Although this finding is sufficient for rejecting our hypothesis that interpersonal and not intrapersonal EC is correlated with individual differences in score estimation, we subsequently tested whether the correlations between *interpersonal* EC and the four individual difference variables differed significantly from the correlations between *intrapersonal* EC and the four individual difference variables. For these tests we used the cocor package in R (via its web interface) for overlapping dependent correlations (Diedenhofen & Musch,

Table 2. Correlation matrix between performance measures (trend = regression slope between mean of the 5 score categories; difference = difference score between the mean of the high lead and far behind category) on the judging score task and measures of emotional intelligence.

	BB_trend	BB_difference	TT_trend	TT_diff	Global_EC	Intra_EC	Inter_EC
BB_trend	–	.89** [.87, .91]	.43** [.34, .51]	.47** [.39, .55]	.11* [.01, .21]	.16** [.06, .26]	.03 [–.07, .13]
BB_difference	–	–	.47** [.39, .55]	.53** [.45, .60]	.13* [.03, .23]	.18* [.08, .28]	.04 [–.06, .14]
TT_trend	–	–	–	.85** [.82, .88]	.08 [–.02, .18]	.10* [.00, .20]	.04 [–.06, .14]
TT_diff	–	–	–	–	.09 [–.01, .19]	.16** [.06, .26]	.00 [–.10, .10]

Note. * $p < .05$; ** $p < .01$. $N = 379$ for all cells. Numbers in brackets indicate the lower and upper limits of 95% confidence intervals.

2015). We used two-sided tests with $\alpha = .05$. Out of the four tested differences ($r_{\text{BB_trend, Intra_EC}}$ VS. $r_{\text{BB_trend, Inter_EC}}$; $r_{\text{BB_difference, Intra_EC}}$ VS. $r_{\text{BB_difference, Inter_EC}}$; $r_{\text{TT_trend, Intra_EC}}$ VS. $r_{\text{TT_trend, Inter_EC}}$; $r_{\text{TT_difference, Intra_EC}}$ VS. $r_{\text{TT_difference, Inter_EC}}$), three proved to be statistically significant ($p = .010$; $p = .010$; $p = .002$): Only when interpersonal differences regarding score estimation in the table tennis videos were assessed by the variable trend (i.e., $r_{\text{TT_trend, Intra_EC}}$ VS. $r_{\text{TT_trend, Inter_EC}}$), we did not find a difference between the correlations with intrapersonal and interpersonal EC ($p = .240$).

Table 3 gives more detailed analyses of the correlations between the ten subscales of the PEC and performance on the score estimation task. The significant correlation between the intrapersonal EC and score estimation performance was mainly explained by how well people described themselves to be able to identify their own emotions, understand their own emotions and regulate their own emotions. The only significant correlations for interpersonal ECs were within the regulation of others' emotions subscale.

Discussion

One goal of Study 1 was to test and replicate both the internal validity and factor structure of the German version of the PEC. Regarding this goal, the German version of the PEC appeared acceptable. The second goal was to test if *interpersonal* EC is associated with an individual's ability to draw accurate inferences from athletes' NVBs during sport competitions. In this respect, the study sought to provide concurrent validity for the important distinction within the PEC between *interpersonal* and *intrapersonal* EC (Brasseur et al., 2013). The pattern of results did not confirm our hypothesis that *interpersonal* EC would be correlated with accuracy of score estimates based on thin-slices of athletes' NVB. Instead, and somewhat surprisingly, *intrapersonal* EC was positively correlated with the score estimation task. Importantly, correlations between *intrapersonal* EC and accuracy were significantly larger than correlations between *interpersonal* EC and accuracy three out of four times, thus suggesting that there is a difference between the predictive quality of *intrapersonal* and *interpersonal* EC (see Diedenhofen & Musch, 2015, for the necessity of directly testing correlations against each other instead of simply comparing significant to non-significant ones).

What might explain this counterintuitive finding? One explanation might be derived from embodied theories of emotion (Niedenthal, 2007). In his book "The Descent of Man", Charles Darwin (1871, p. 492) wrote: "The sight of another person enduring hunger, cold, fatigue

Table 3. Correlation matrix between performance measures (trend = regression slope between mean of the 5 score categories; difference = difference score between the mean of the high lead and far behind category) on the judging score task and individual emotional intelligence scales.

	ident_own	ident_other	und_own	und_other	expr_own	listen_other	regul_own	regul_other	use_own	use_other
BB_trend	.119*	.04	.146**	.031	.043	-.041	.114*	.078	.063	.011
BB_difference	.119*	.057	.169**	.028	.078	-.042	.147**	.101*	.049	.010
TT_trend	.119*	.049	.045	.037	.043	-.049	.038	.112*	.081	.006
TT_diff	.180**	.036	.123*	.004	.043	-.088	.093	.087	.046	-.014

Note 1 . * $p < .05$; ** $p < .01$. $N = 379$ for all cells.

Note 2. Identify my emotions (ident_own); identify others' emotions (ident_other); understand my emotions (und_own); understand others' emotions (und_other); express my emotions (expr_own); listen to others' emotions (listen_other); regulate my emotions (regul_own); regulate others' emotions (regul_other); use my emotions (use_own); use others' emotions (use_other).

revives in us some recollection of these states, which are painful even in the idea.” This quote by Darwin nicely captures the central premise of embodied theorising: that is, knowledge about the world is grounded in the original neural states that occurred when the information was first acquired. Stated differently, using knowledge is a lot like reliving past experience in at least some of its sensory, motor, and affective modalities (Duckworth et al., 2002; Niedenthal, 2007; Strack et al., 1988). In addition, theorising on mirror neurons (e.g., Gallese & Goldman, 1998) could provide a further line of explanation of how experiencing and observing certain states and/or behaviours rely on shared neuronal mechanisms. In regard to the present findings, this would mean that the correlation between *intrapersonal* EC and accuracy of score estimations might be explained similarly: That participants who described themselves as having high *intrapersonal* EC were better able to reinstate similar experiences they had when they were winning or losing in sports competitions and this helped them to draw more accurate inferences from the observed NVB of athletes. This explanation fits well with a growing body of research that provides evidence for a reciprocal relationship between the bodily expression of emotion and interpreting emotional information: “... experiencing an emotion, perceiving an emotional stimulus, and retrieving an emotional memory all involve highly overlapping mental processes” (Niedenthal, 2007, p. 1003).

Study 2

We have to acknowledge that the correlation between *intrapersonal* EC and the score estimation task was small and emerged coincidentally, opposite to what we had hypothesised. Therefore, we decided to run an additional study that sought to corroborate the surprising finding of Study 1. To this end, we conducted a second study in which participants again completed the German PEC in addition to a validated test assessing individual differences in emotion recognition competence (the Emotion Recognition Index, ERI; Scherer & Scherer, 2011) and a different sports task in which participants had to estimate the score of professional Darts players based on their NVB directly before throwing Darts in a professional Darts tournament (Furley & Memmert, 2021).

Method

Participants

310 participants (170 male and 140 female; $M_{age} = 26.0$ years; $SD = 8.4$ years; three did not indicate their age and for 5 data from the ERI was missing due to a technical error) took part in the study. Every participant gave written informed consent. Sample size considerations were similar to Study 1 and the sample had sufficient power (.95) to detect $r = .20$ on two-tailed Pearson correlations (Faul et al., 2007). Again, the majority of participants had no particular sports expertise and were college students enrolled at German universities, primarily in sport science or psychology majors. All participants participated voluntarily without compensation and were recruited within research methodology classes and bachelor and master's projects.

Profile of Emotional Competence

For the assessment of EC we used the same German version of the Profile of Emotional Competence (PEC; Brasseur et al., 2013) as in Study 1.

Emotion Recognition Index (ERI)

The full details of the ERI can be found in Scherer and Scherer (2011). It is considered one of the few validated tests to assess individual differences in emotion recognition competence. It is a rapid (approximately 10–15 min) instrument that consists of a facial and a vocal subtest that both measure accuracy in discriminating emotion expressions in the face and in the voice. For the facial test a subset of facial images from Ekman and Friesen's (1976) stimulus material was presented for 3 s and participants were asked to indicate which emotion is being expressed in a multiple-choice quiz with the options of anger, fear, joy, sadness, and disgust (without time limitation). For the vocal subtest recordings of radio actors were used that are part of a large database of emotional audio recordings (GVEESS, Banse & Scherer, 1996). These recordings resemble natural speech, however without meaningful content. After playing the approximately two-seconds-recordings, participants were again asked to indicate which emotion was being expressed in a multiple-choice quiz with the options anger, fear, joy, sadness and neutral (without time limitation). Participants started with the facial test followed by the vocal test. Each subtest started with two examples followed by the 30 items respectively. The test procedure was computer-based and instructions were presented on a computer screen. Participants were asked to respond in a fast and spontaneous manner. All data were stored and analyzed automatically via Limesurvey (URL <http://www.limesurvey.org>). A response corresponding with the target was scored as 1 if correct and as 0 if incorrect. A total facial and vocal score were computed from the number of items in which the participant's response matched the target emotion. High scores represented good emotion recognition skills.

Predicting performance in televised sports competitions

Televised video recordings of almost all matches during the 2017 Professional Darts Corporation (PDC) World Darts Championship were sampled (see Furley & Memmert, 2021 for full details on task and selection procedure). This procedure gave us video footage of 36,168 individual dart scores (12,056 throws) of 61 of the 64 players.

For the random selection procedure applied in this test, we created four different performance categories for every throw (i.e., three darts): poor performance for professional Darts players was categorised if none of the three darts hit a high triple segment (i.e., triple-20; triple-19, triple-18, and triple-17). Medium performance is if one of the three darts of a throw hits a high triple segment. Good performances were if 2 out of three darts hit high triple segments. Excellent performance was if all three darts hit high triple segment. Importantly, we only categorised darts in which the players were aiming for triples as the final darts in a leg have to hit a specific double field to finish the leg or are thrown to targets to set up a preferred double segment that has to be hit with the final dart.

Every throw in the four respective performance categories was serially numbered. Then we created 20 random numbers (in the respective number ranges within the performance categories) via the page random.org in the four performance categories and cut the selected throws (i.e., three individual clips of pre-performance NVB of the respective darts in a throw; each video showed the pre-performance nonverbal behaviour of three individual darts in successive order) with the software Adobe Premiere 4.0. The

videos consisted of a resolution of either 720×1280px with a frame rate of 30.00 fps (frames per second) or 360×640px with a frame rate of 29.94fps depending on the type of video files that were available to us. A throwing sequence was determined as valid, if the dart player was shown in a close-up before and during all three darts in the randomly selected throw. Therefore, the starting point of a throwing sequence was selected by the first frame of which a movement of the throwing arm appeared to prepare and initiate the throw (e.g., when a player started to lift the hand of his throwing arm from a resting position, by holding the hand in front of his body). The last frame before the dart left the hand of the player was defined as the ending point of the throwing sequence. Selected throwing sequences were dismissed, if the sequence was either too short (i.e., < 1 s) or the player was not shown during the throw. As a replacement of the dismissed throw the following throw in the corresponding list for each performance category was used (e.g., if throw 1134 was selected in the poor performance category, but was missing a close-up of the pre-performance NVB of the second dart, then throw 1135 was included as stimulus material in the study). The mean duration of all videos was 3.9 sec ($SD = 1.1$ sec). The videos did not significantly differ in length between the experimental categories ($p = .235$ in 1×4 [number of triples scored] univariate ANOVA on the length of videos).

The test was programmed with PsychoPy (Meinold & Furley, 2018; Peirce & MacAskill, 2018). The respective software randomly sampled 10 videos from each of the four performance categories and showed the 40 selected video clips in different random orders for every participant. This approach helps to ensure that results do not depend on specific combinations of stimuli. After every video clip participants were asked to estimate the score of all *three* darts of the throw for which they had just seen the pre-performance NVBs. All videos were presented silently to ensure that ratings were based on NVB and not, for example, crowd noise. Participants gave their ratings by moving a mouse cursor from the middle (90 points) of a 180 points scale to either the left pole (0 points; poorest possible performance) or right pole (180 points; highest possible performance) and clicking the left mouse button to log in their rating for the respective throw.

Preliminary study

To verify that the videos in the four performance categories differed in terms of emotional content we again conducted a preliminary study in which participants were asked to give their gut judgment if the displayed Darts player was in a positive or negative emotional state by pressing the left-arrow key for a negative state and the right-arrow key for a positive state. We decided for this type of manipulation check given the lack of an effect of emotional intensity in the manipulation check measure of Study 1. The videos were played in a random order and after the video had finished, participants had 2 s to give their judgment.

50 participants ($M_{age} = 28.3$; $SD = 11.0$; 19 female) rated the 80 Darts videos in terms of whether the displayed athlete was in a positive or negative state prior to his performance. A 1×4 (number of triples scored) repeated measures ANOVA on the cumulated positive and negative ratings of the respective triple categories revealed a significant main effect of performance category (0-triple, 1-triple; 2-triple; 3-triple) on the emotional valence ratings towards the target players in the study ($F(3, 147) = 8.31$, $p < .001$, $\eta_p^2 = .15$). Participants ratings showed a linear trend as a function of performance category

with more negative emotional ratings in the lower performance categories and more positive emotional ratings in the higher performance categories. Therefore, the videos seem suitable to test if emotional competencies as measured with the PEC are predictive of drawing accurate inferences from the NVBs of athletes.

Procedure

Participants filled out a questionnaire gathering demographic data at the start of the experiment. They were tested individually on a standard 17 inch notebook placed approximately 60 cm away from them. Participants were informed about the purpose of the experiment at the end of the procedure.

Data analyses

First, we analyzed the overall effect of how well participants could predict the performance of professional Darts players based on thin-slices of preparatory NVB with a 1×4 (poor, medium, good, excellent) repeated measures ANOVA on the average score estimates within the respective experimental categories. We followed up this ANOVA with a polynomial linear contrast analysis.

In order to analyze individual difference in any potential overall effects, we again computed two variables as an index of how well participants were able to predict the score of the observed athletes. For the first individual difference variable we simply subtracted the average scores in the poor-performance-category from the average score in the excellent-performance-category. This variable was labelled *difference*. Higher differences in this variable indicate better performance in the score prediction task. We computed a further variable that included the means of all four performance categories as a potentially more sensitive individual difference variable in score-prediction performance. To this end we calculated the slope of the regression line of the four score categories for every participant using the same formula as in Study 1. This variable was labelled *trend*. The x-values for every participant were set as 1 for poor performance, 2 for medium, 3 for good performance, 4 for excellent performance. Again, lower values for the regression slope indicate poorer performance on the score prediction task and higher values indicate better performance. To test for an association between the different EC as measured with the PEC and the performance on the score estimation task we ran a series of two-tailed Pearson correlations and tested them against each other.

Results

The descriptive results of Study 2 are displayed in [Figure 2](#). A 1×4 (number of triples scored) repeated-measures ANOVA on the mean estimated scores of the respective triple categories revealed a significant main effect of triple category (0-triple, 1-triple; 2-triple; 3-triple) on the estimated scores of the target players ($F(3, 927) = 61.55, p < .001, \eta_p^2 = .17$). Polynomial linear contrast analysis showed a strong linear relationship between the score estimates and the score categories, $F(1, 309) = 145.55, p < .001, \eta_p^2 = .32$. In this respect, the findings replicate the results from Furlley and Memmert (2021).

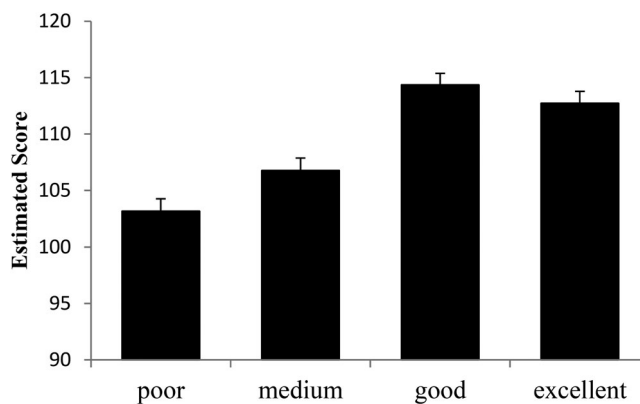


Figure 2. Mean score predictions as a function of the four performance categories. Error bars represent standard errors.

We consider it important to mention that our participants had little experience regarding Darts and domain-specific experience in either playing or watching Darts was neither significantly correlated with the difference nor the trend measures in the thin-slices task (all r between $-.057$ and $.077$, and $p > .190$). We consider this important as it rules out an alternative explanation of the results: That participants recognised some of the famous and dominating Darts players and inferred that these are more likely to perform well and coincidentally the applied random selection procedure led to better players being more prevalent in the better performance categories (when testing for this we did not find that player ranking significantly differed across the four performance categories).

Individual differences in predicting the score based on athletes' NVB

Table 4 shows that both individual difference variables in the score estimation task (difference and trend) were not significantly correlated with any of the EC measures from the PEC. Hence, in contrast to both our original hypothesis, and our hypothesis derived from study 1, EC score was not significantly correlated with performance in the score-prediction task. Stated differently, people with better self-reported EC were not better at drawing inferences from NVB of professional athletes during sport competitions. This was also not the case when factoring in domain-specific experience in either playing or watching Darts in partial correlations between the EC measures and the difference or the trend measures in the thin-slices task (all r between $-.05$ and $-.01$, and $p > .369$). Likewise, both individual difference variables in the score estimation task (difference and trend) were not significantly correlated with any of the ERI scores.

Importantly, both the ERI_{face} and the ERI_{voice} were significantly correlated with the *interpersonal* EC, but not the *intrapersonal* EC. This result is in contrast to the results of Study 1 that showed correlations between *intrapersonal* EC and the thin-slices sports task. We again tested whether the correlations between *interpersonal* EC and the ERI variables differed significantly from the correlations between *intrapersonal* EC and the ERI variables using the same procedure as in Study 1. Out of the two tested differences ($r_{ERI_{face}, Intra_{EC}}$ vs. $r_{ERI_{face}, Inter_{EC}}$; $r_{ERI_{voice}, Intra_{EC}}$ vs. $r_{ERI_{voice}, Inter_{EC}}$), only the one for the ERI_{voice} proved to be statistically significant ($p = .030$), while ERI_{face} failed to reach the significance threshold ($p = .060$).

Table 4. Correlation matrix between performance measures (trend = regression slope between mean of the 4 performance categories; difference = difference score between the mean of the poor and excellent performance category) on the score prediction task and measures of emotional competences.

	Darts_trend	Darts_difference	ERI_face	ERI_voice	Global_EC	Intra_EC	Inter_EC
Darts_trend	–	.95** [.94, .96]	–.07 [–.18, .04]	–.02 [–.13, .10]	–.03 [–.14, .09]	–.02 [–.13, .10]	–.03 [–.14, .08]
Darts_difference	–	–	–.09 [–.20, .03]	–.06 [–.17, .06]	–.04 [–.15, .08]	–.04 [–.15, .08]	–.03 [–.14, .08]
ERI_face	–	–	–	.28** [.17, .38]	.15** [.04, .26]	.08 [–.03, .19]	.18** [.07, .29]
ERI_voice	–	–	–	–	.09 [–.02, .20]	.02 [–.09, .13]	.14* [.03, .25]

Note. * $p < .05$; ** $p < .01$. $N = 379$ for all cells. Numbers in brackets indicate the lower and upper limits of 95% confidence intervals.

Table 5 gives more detailed analyses of the correlations between the ten subscales of the PEC and performance on the score prediction task and both of the ERI scores. The significant correlation between the *interpersonal* EC and the *ERI_face* was mainly explained by how well people described themselves to be able to identify others' emotions and understand others' emotions. The significant correlation between the *interpersonal* EC and the *ERI_voice* was mainly explained by how well people described themselves to be able to understand others' emotions and regulate others' emotions.

Discussion

The goal of Study 2 was to corroborate the surprising findings from Study 1 that *intrapersonal* EC instead of *interpersonal* EC was associated with emotion recognition competence and using this competence for drawing inferences when observing NVB in sports competitions. In this respect, the pattern of results of Study 2 was clear as there was absolutely no indication that *intrapersonal* EC was either associated with better ERI performance in the face or in the voice or with drawing more accurate inferences from athletes' NVB during sport competitions. In this respect, we did not find evidence for the embodied theorising on emotions (Niedenthal, 2007) that was used in explaining the results from Study 1. Generally speaking, these diverging patterns of results support the necessity of replicating effects found with one set of stimuli or tests with different stimuli/tests to verify that the phenomenon of interest generally applies and is not dependent on a highly specific set of stimulus material or tests (e.g., Fiedler, 2011). The sensitivity of conclusions drawn from individual studies using specific tests or paradigms is further highlighted by the finding that the *interpersonal* PEC score correlated with the *ERI_face* and *ERI_voice* score. This result indicates that self-reported *interpersonal* EC was associated with a better competence of recognising (exaggerated) emotional expressions by actors. However, we have to acknowledge that the correlations were small.

General discussion

Given the overwhelming interest and popularity, but also controversy on assessing EI (e.g., Brasseur et al., 2013; Goleman, 1995; Matthews et al., 2007), the goal of the

Table 5. Correlation matrix between performance measures (trend = regression slope between mean of the 4 performance categories; difference = difference score between the mean of the poor and excellent performance category) on the score prediction task and the individual emotional competences scales.

	ident_own	ident_other	und_own	und_other	expr_own	listen_other	regul_own	regul_other	use_own	use_other
Darts_trend	−.007	−.039	.072	−.058	−.030	−.012	.028	−.036	−.005	.036
Darts_difference	−.041	−.036	.063	−.041	−.024	−.015	−.007	−.046	.011	.026
ERI_face	.086	.145*	−.004	.202**	.101	.102	.006	.092	−.005	.100
ERI_voice	.041	.101	−.028	.142*	.037	.059	.021	.117*	−.041	.083

Note 1. * $p < .05$; ** $p < .01$. $N = 310$ for all cells except $N = 305$ for the cells involving the ERI scores.

Note 2. Identify my emotions (ident_own); identify others' emotions (ident_other); understand my emotions (und_own); understand others' emotions (und_other); express my emotions (expr_own); listen to others' emotions (listen_other); regulate my emotions (regul_own); regulate others' emotions (regul_other); use my emotions (use_own); use others' emotions (use_other).

present research was to validate a recently emerged measurement tool that distinguishes between different EC (i.e., identifying, expressing, understanding, regulating, using), at the *interpersonal* and *intrapersonal* levels (PEC; Brasseur et al., 2013). In this respect we argued that sports competitions would be a suitable context to seek concurrent validity for the *interpersonal* dimension of the PEC given the prevalence of naturally occurring emotional expressions in sports (see Furley & Schweizer, 2020 for a review). More specifically, in two high-powered studies we tested if *interpersonal* EC (as opposed to *intrapersonal* EC) were predictive of interpreting the nonverbal expressions of others. Study 1 did not confirm our initial hypothesis that *interpersonal* EC would be correlated with accuracy of score estimates based on thin-slices of athletes' NVB. Instead, and somewhat surprisingly, *intrapersonal* EC was correlated (albeit weakly) with the score estimation task. We speculated that embodied theories of emotion (Niedenthal, 2007) might be able to account for the surprising finding that participants who described themselves as having high *intrapersonal* emotional competence were better able to reinstate similar experiences they had when they were winning or losing in sports competitions and this helped them to draw more accurate inferences from the observed NVB of athletes. However, when we tried to replicate this coincidental finding in Study 2, we found no indication that *intrapersonal* EC was correlated with drawing more accurate inferences from athletes' NVB during sport competitions. Neither was *intrapersonal* EC correlated with performance in a validated test assessing individual differences in emotion recognition competence (the Emotion Recognition Index, ERI; Scherer & Scherer, 2011). However, *interpersonal* EC score was correlated (albeit weakly) with the ERI_{face} and ERI_{voice} score.

What does this pattern of results tell us about the predictive validity of the PEC measure of EC? First, we want to point out that both Study 1 and Study 2 were highly powered (Schweizer & Furley, 2016) which increases our confidence that null findings are in fact null findings and decrease the chances that significant correlations are inflated. Therefore, we can conclude that self-reported EC measured with the PEC are not predictive of drawing inferences from naturally occurring nonverbal displays in sports. One explanation for this may be related to the tripartite model of emotional intelligence (Mikolajczak, 2009), assuming that beyond the *intrapersonal* and *interpersonal* aspects of EC, three stratified levels have to be considered regarding EC: knowledge, ability, and trait. Knowledge reflects what one knows about EC, ability reflects the EC related techniques an individual is able to use when prompted to do so, while the trait level reflects what one typically does when facing situations involving EC. In the current study, EC were assessed via self-report measurements, which most likely corresponds to the trait level. However, the tasks used in studies 1 and 2 were closer to the performance tests used to measure ability EC, like the Mayer-Salovey-Caruso Emotional Intelligence Test (Mayer et al., 2002). However, in Study 2 we also did not find an association between the ERI (Scherer & Scherer, 2011) and the sports thin-slices task. Further research including similar tasks should therefore test comparatively the predictive power of ability and trait EC measurement tools, given they reflect different facets of EC.

The present research shows that further work is needed regarding the validity of measures assessing *interpersonal* emotional competencies which are no doubt important in various sport performance settings (e.g., Laborde et al., 2017). Within the current debate on the crisis of confidence in psychology, which mainly refers to the fact that

many findings within psychology do not replicate (Camerer et al., 2018; Open Science Collaboration, 2015), it has further been stated that psychology suffers from a validation crisis (Schimmack, 2010; Schimmack, 2019). In this respect, Schimmack (2019) has noted that everybody would probably agree that good science requires valid measures. However, a critical assessment of the psychological literature indicates that many highly popular and frequently used measures in psychology have not been validated adequately (Schimmack, 2019). This argumentation is, for example, supported by a recent highly powered study (overall $N = 2,641$) comparing various measures (the Self-Control Scale, the Stroop, and Flanker task) that all have been used to measure the construct of self-control (Saunders et al., 2018). The results suggested little-to-no relationship between self-reported self-control and performance on the Stroop and Flanker tasks, which have been argued to measure the same psychological constructs. Similarly, different measures that are again proposed to assess the same psychological constructs, in this case implicit motives of a person, showed either no or very little correlation (Schüler, Brandstätter, Wegner, & Baumann, 2015). Unfortunately, the results of the present research in combination with previous research on the measurement of EI (e.g., Elfenbein & MacCann, 2017; Laborde & Allen, 2016; Olderbak et al., 2019), point in a similar direction and suggest that the psychological construct of EI or EC is not only difficult to measure, but also has multiple dimensions, and limited predictive validity to situations it arguably claims to be predictive of (Matthews et al., 2004).

The present research approach has some notable strengths and weaknesses. We consider it a strength that our stimulus materials were derived from actual sports competitions, instead of being artificially created, which is often the case within research in EI (Elfenbein & MacCann, 2017; Olderbak et al., 2019; Scherer & Scherer, 2011). Nevertheless, a limitation of the present research is that the emotional content shown in the sports clips was rather subtle and it therefore remains unclear if the measured ECs might be better predictive of distinguishing between stronger emotional expressions in sports. Further research might want to investigate into this possibility. Another limitation of the present research is the correlational nature of the studies and it is important to not mistake “absence of evidence” for “evidence of absence”. In the present research, we only used two measures of EC and correlated these with two tasks that we have used in previous research that arguably require ECs to be performed successfully (i.e., drawing inferences based on NVB in sports). As stated before, we therefore recommend that further research should look into the link between other measures of EC and emotions in sports, like the Mayer-Salovey-Caruso Emotional Intelligence Test (Mayer et al., 2002).

In conclusion, The present research addresses the increasing calls for ways of increasing construct validity of popular psychological constructs (Eronen & Bringmann, 2021; Flake et al., 2017) as popular self-report measures of EI lack high-powered validation studies and might not measure what they claim to measure. Stated differently, even if people report themselves to be skilled at identifying and understanding other people’s emotional states (which is an important skill in the context of sports), they might not be as skilled at this feat as they report themselves to be if tested in a performance task. Although, both the internal validity and factor structure of the German version of the PEC appear acceptable, the present findings appear problematic concerning the divergent validity of the *inter*- and *intrapersonal* EC subscales. Granting that measures

of EI have face validity in the field of sports, the present research highlights the necessity of rigorous validation studies before measures like the PEC are used in sport psychological research and application.

Research Disclosure statements

The authors declare that 1) (a) the total number of excluded observations and (b) the reasons for making these exclusions have been reported in this manuscript; 2) that all independent variables or manipulations, whether successful or failed, have been reported in the manuscript; 3) that all dependent variables or measures that were analyzed for this article's target research question have been reported in the manuscript/online supplements; 4) that (a) how sample size was determined and (b) our data-collection stopping rule have been reported in this manuscript.

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All data will be made available at reasonable request to the corresponding author.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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